Vamsi Alla

Senior Software Engineer · Remote, USA · vamsialla.dev@gmail.com · vamsialla.com · linkedin.com/in/vamsi-alla · github.com/vamsialladev

SUMMARY

12+ years building production software, the last 3+ designing, evaluating and deploying LLM and agentic AI applications for enterprise clients. I work directly with client engineers, product owners and executives to find the real problem, then own the solution end to end, from architecture and retrieval to evaluation, deployment and handoff. Open to Forward Deployed Engineer and senior AI engineering roles.

SELECTED IMPACT

- Cut document processing time by ~60% and model running cost by ~55% by replacing a manual enterprise workflow with a multi-agent AI platform I architected and delivered end to end.
- Held 90%+ field accuracy on structured extraction in production by routing low-confidence fields to human review instead of letting the model guess.
- Built a reusable RAG layer, evaluation harness and guardrail configurations that later engagements adopted as their starting point instead of rebuilding.
- Tuned vector retrieval and metadata filtering in Aurora PostgreSQL with pgVector to interactive-speed semantic search across a large job taxonomy.

EXPERIENCE

Senior Software Engineer · Apr 2023 – Present · MI, USA · Remote

- *Enterprise Multi-Agent AI Platform* (Aug 2025 – Present): own the architecture of an AI platform on AWS that automates multi-step document work; specialized agents under a supervisor over A2A and a shared MCP layer; RAG over FAISS and pgVector; serverless APIs (Lambda, Step Functions, API Gateway); guardrails, PII protection and prompt-injection defense with RAGAS and regression suites; a model-agnostic interface (Bedrock, OpenAI, Anthropic) with prompt caching and right-sized endpoints. Work directly with client stakeholders from engineers to executives.
- *Document Workflow Orchestration* (Sep 2024 – Aug 2025): unstructured documents to schema-aligned JSON with GPT-4o and Pydantic validation; explicit "unknown / needs confirmation" behavior instead of fabricated fields; FastAPI services built with client engineers.
- *Job Semantic Search Platform* (Feb 2024 – Aug 2024): embeddings ingestion and semantic search APIs on Aurora PostgreSQL with pgVector; monitoring for ingestion, latency and relevance; team standards for RAG and evaluation.
- *Travel Planning Copilot & Virtual Concierge* (Apr 2023 – Jan 2024): customer-facing assistant with schema-driven itineraries and Google Places; React chat front end; clarification, safe refusal and escalation paths; retries, timeouts and regression prompt suites.

Java Developer, Enterprise Microservices Platform · TIAA · Jul 2022 – Mar 2023 — Spring Boot APIs and microservices, zero production incidents, drove SonarQube adoption.

Java Developer, Financial Change Management · Wells Fargo · Feb 2021 – May 2022 — Spring Boot services with US counterparts, upstream and downstream issues resolved within SLA.

Java Developer, Product Data Management · Client: Apple · Feb 2015 – Jan 2021 — Spring MVC and Spring Boot applications across the full SDLC, REST and SOAP services, Hibernate.

Java Developer, SOA Integration & Healthcare Systems · Napier · Feb 2014 – Jan 2015 — SOA integrations and REST services for telecom and healthcare clients, including Telstra's authentication and authorization API.

SELECTED PROJECTS

Independent, simulated client engagement for a fictional hotel group, synthetic data · vamsialla.com/projects

- **Resort Activities Concierge** (agent + RAG + Postgres, staff approval): 100% of checks on 61 test cases × 3 runs at $0.76 per 1,000 messages · github.com/vamsialladev/resort-activities-concierge
- **Guest Request Agent** (six typed tools, staff approve every reply): 100% escalation recall and injection resistance, CI-gated, at $1.24 per 1,000 emails · github.com/vamsialladev/guest-request-agent
- **Eval Harness** (one regression gate for two AI apps): caught a deliberate top-k 5 → 2 regression that a confidence interval alone missed · github.com/vamsialladev/eval-harness
- **Hotel Operations Knowledge Assistant** (cited RAG, live): 100% answer accuracy on a 54-question eval at $0.18 per 1,000 questions · github.com/vamsialladev/hotel-ops-assistant

SKILLS

AI: Claude, GPT-4o, Bedrock, Azure OpenAI; LangChain, LangGraph, CrewAI; multi-agent, tool calling, MCP, A2A, structured outputs · Retrieval & evals: hybrid search, pgVector, FAISS, Pinecone; RAGAS, LangSmith, regression suites; guardrails, PII redaction, prompt-injection defense · Backend: Python, Java, SQL, TypeScript; FastAPI, Spring Boot, Kafka; React, Streamlit · Cloud: AWS (Lambda, Step Functions, EKS, SageMaker, Bedrock), Azure, GCP; Docker, Kubernetes, Terraform, GitHub Actions

EDUCATION

Master of Computer Applications (MCA) and Bachelor of Science (B.Sc.), Acharya Nagarjuna University, India